

P13: Building a Bee Monitoring Pipeline for Environmental Observation

Group Members:

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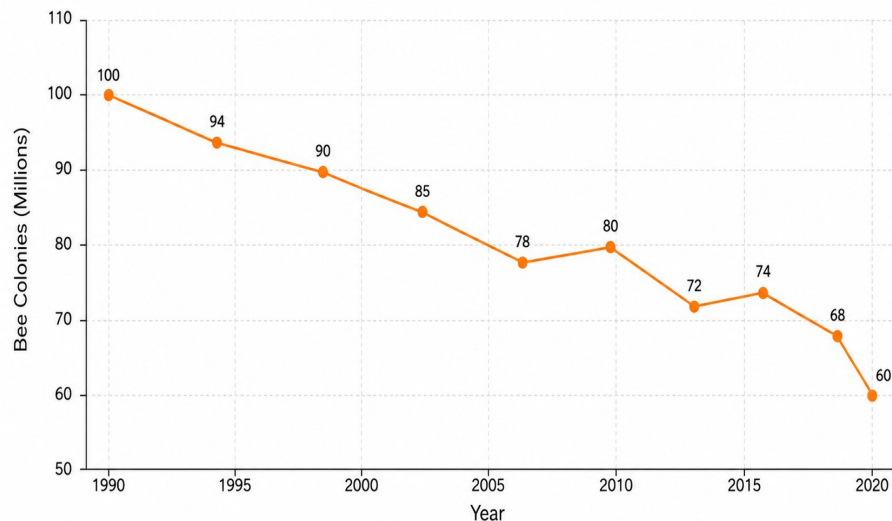
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Lennart Reißner**

Motivation



Bee Population Decline (1990–2020)

Estimated global managed honey bee colonies (in millions)



- Importance of Pollinators in biodiversity and food production.
- Decline in Bee Population in last few decades.
- Challenges in Traditional Monitoring.
- Why Computer Vision?

Problem

Small and fast-moving targets

Challenging visual conditions



No scalable pipeline exists

Limited annotated data

Primary Goal: Monitoring Pipeline



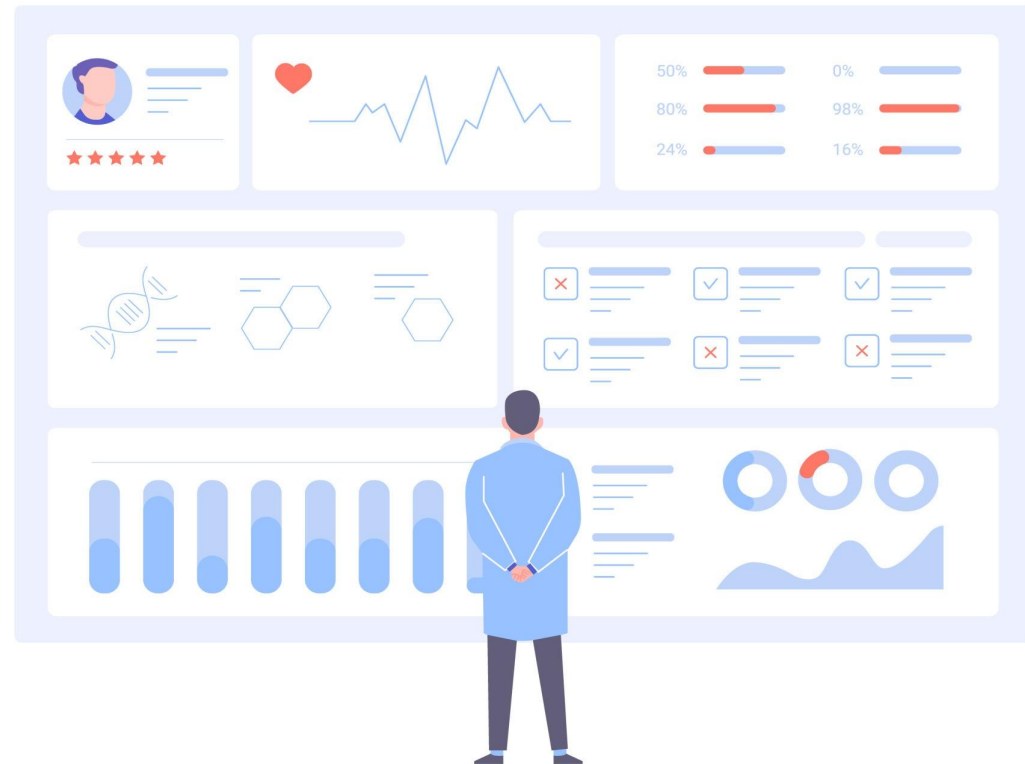
1. Model Development

Teaching the system to see and track bees automatically.



2. Live Insights

Translating data into clear, interactive visual dashboard.



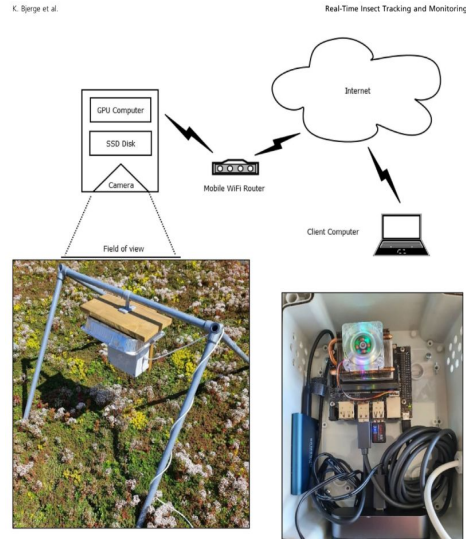
Our Approach

- **Step 1 - Dataset Creation:** Extract frames from bee videos using OpenCV, annotate with CVAT, split into train/val/test sets.
- **Step 2 - Data Augmentation:** Apply geometric and color augmentations using Albumentations; generate synthetic images.
- **Step 3 - Detection Model:** Train YOLOv 8 on the bee dataset; evaluate using Precision, Recall, and mAP metrics.
- **Step 4 - Tracking:** Implement DeepSORT or ByteTrack to maintain bee identities across frames.
- **Step 5 - Pipeline and Dashboard:** Integrate all modules; build Streamlite dashboard showing bee counts, heatmaps, and temporal trends.

Current Progress

- ML environment and Bender cluster configured.
- Initial literature review on bee detection and tracking completed.
- Studied object detection approaches (YOLO v8, tracking methods, SAHI).
- Explored strategies for small object detection and tracking enhancement.

Literature Review



Bjerge et al. (2021)

- Real-time insect tracking and monitoring

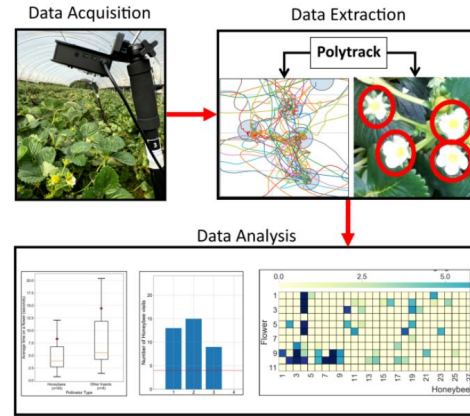


Figure 2: Proposed pipeline for computer vision-based pollination monitoring

Ratnayake et al. (2021-2022)

- Pollination monitoring using CV and deep learning.

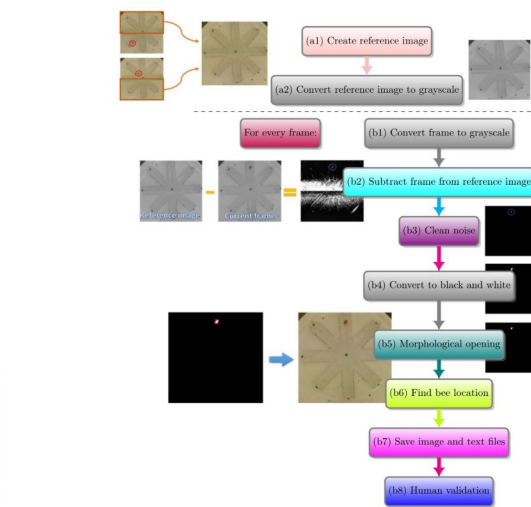
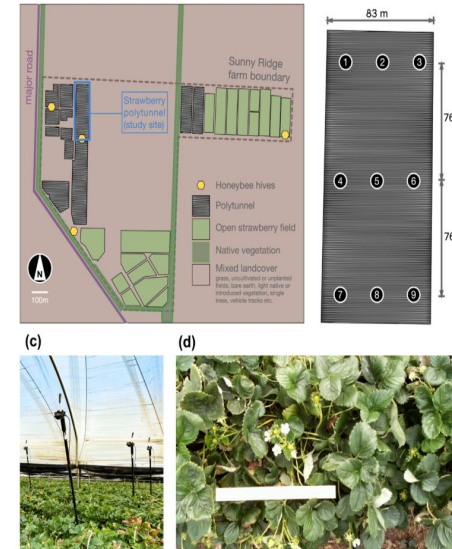


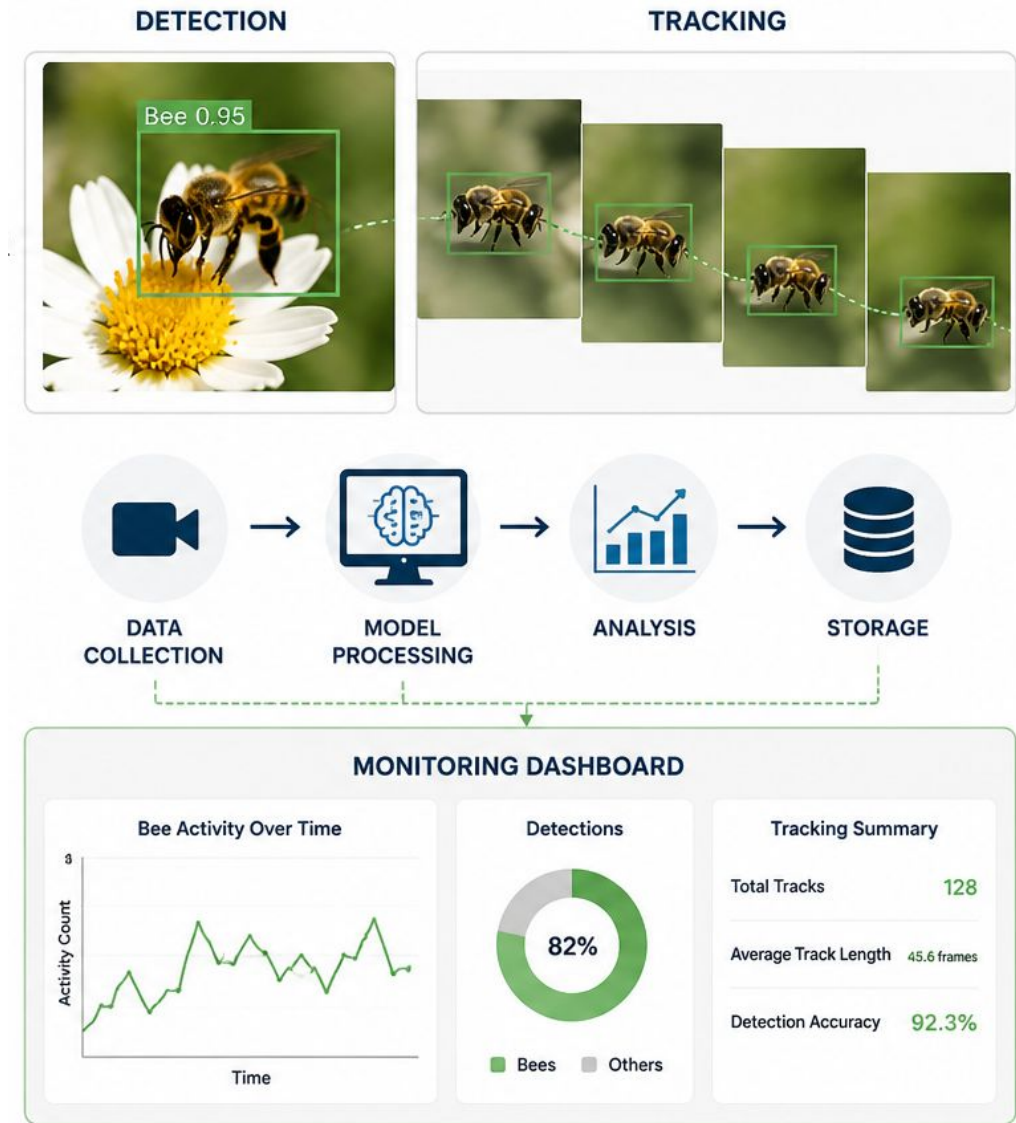
Fig. 5 Preparing the training dataset for the planar bee tracking system model

Rozenbaum et al. (2024)

- Machine learning-based bee recognition and tracking.

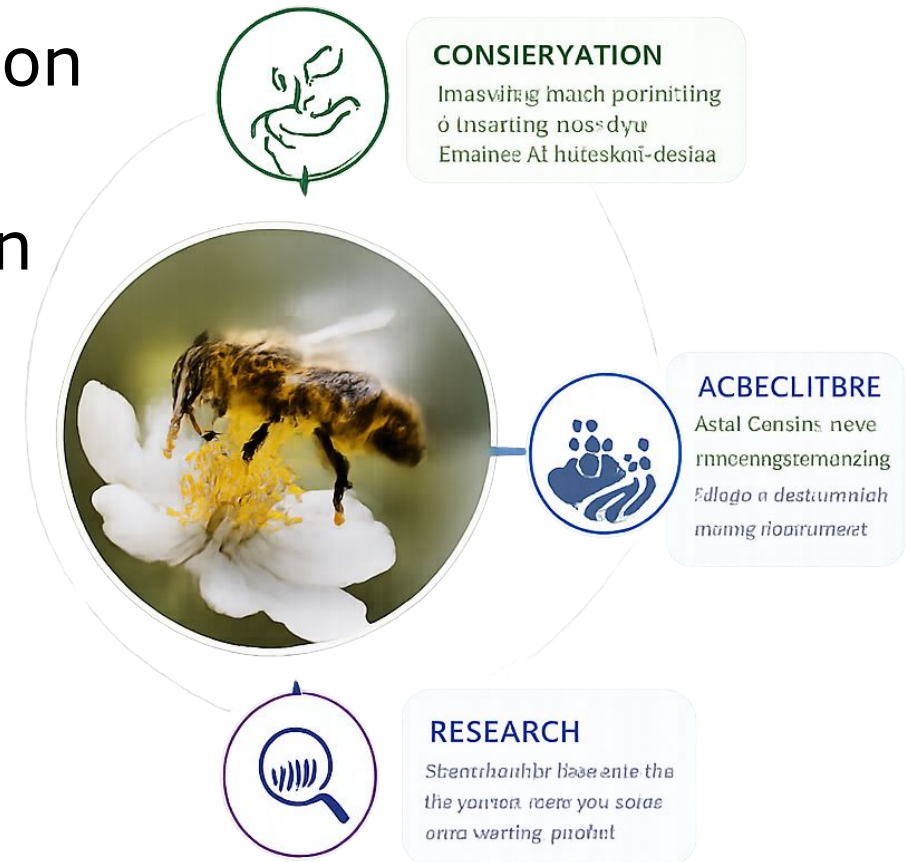
Expected Results

- Robust Bee Detection Model
- Reliable Tracking System with algorithm
- Improved Detection
- Monitoring Pipeline
- Performance Evaluation



Impact

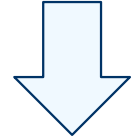
- **Conservation:** Supports continuous pollinator monitoring and early detection of bee population decline.
- **Agriculture:** Helps improve pollination monitoring and provides insights into bee activity in crop environments.
- **Research:** Provides a reusable framework for future bee detection, tracking, and behavioural analysis studies.



Next Steps

- **Dataset Improvement** : Refine annotations and improve dataset quality for better robustness under different conditions.
- **Model Experimentation** : Train and compare different detection models (YOLOv8 n / s / m / l) to identify the best-performing approach.
- **Detection Optimization** : Fine-tune training parameters (epochs, image size, augmentation, hyperparameters) to improve accuracy.
- **Tracking Integration** : Implement and evaluate tracking methods (ByteTrack / DeepSORT) for bee movement analysis.
- **Performance Evaluation** : Compare models using Precision, Recall, mAP, and tracking consistency.

Better dataset



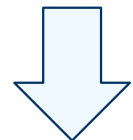
Try models



**Improve
detection**



Add tracking



Evaluate

Summary

Purpose: Bees are essential for growing the food we eat, but they are currently in danger. Because it is too slow and difficult for humans to watch every bee by hand, we need a smarter way to protect them.

Method: We created a computer system that acts like a 24/7 digital observer. It uses cameras and artificial intelligence to automatically recognize and follow individual bees as they fly, doing the work that humans used to do manually.

Result: This system turns video into simple charts and maps. These visuals show exactly where and when bees are most active, giving farmers and scientists the data they need to keep bee populations and our food supply healthy.

Project Plan

Project Plan: Key Achievements Reached

- Goal: Established a baseline bee detection pipeline through dataset understanding, model research, and initial training and Explored detection and tracking approaches to improve monitoring performance in future phases.
- Plan update: Initial focus shifted toward improving dataset quality and detection performance before implementing advanced tracking methods to ensure a stronger foundation.

Tasks of Divyesh (s54dkana)	Tasks of Prince (s57pbamb)	Tasks of Aniket (s17alimb)	Tasks of Rohit (s09rrath)
- Basic understanding of the dataset - Research on model training and bee detection - Built a base-level bee detection model - Studied YOLO architectures for object detection	- Literature review on bee detection and tracking methods - Compared YOLOv5, YOLOv7, and YOLOv8 models - Researched DeepSORT and ByteTrack tracking algorithms	- Studied environmental monitoring systems - Setup Python, OpenCV, and Jupyter environment - Worked on evaluation metrics and research documentation	- Studied insect and pollinator monitoring datasets - Researched and compared detection models for bee monitoring

Project Plan: Key Achievements Planned

- Goal: Improve bee detection accuracy through dataset refinement and model optimization and Begin integrating robust tracking methods for reliable pollinator monitoring

Tasks of Divyesh (s54dkana)	Tasks of Prince (s57pbamb)	Tasks of Aniket (s17alimb)	Tasks of Rohit (s09rrath)
- Improve model accuracy using advanced methods - Implement class balancing techniques - Experiment with different detection algorithms - Optimize bee detection for small objects	- Apply data augmentation using Albumentations - Prepare train/validation/test datasets - Integrate DeepSORT and ByteTrack tracking - Improve video processing pipeline	- Analyze tracking performance and movement patterns -helps to prepare datasets - Research real-time deployment possibilities - Work on environmental analytics integration	- Design and develop the Streamlit monitoring dashboard - Implement temporal bee activity charts - Write and compile the final project report